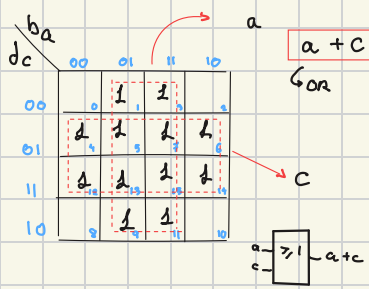
Boletín nº 7

Sistemas Digitales: Boletines

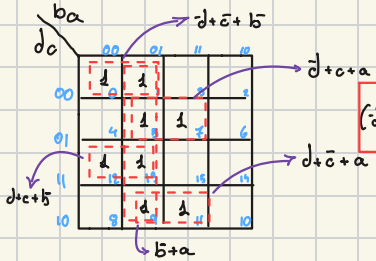
Ejercicio 1:

Simplifica las siguientes funciones utilizando el método de *Karnaugh-Veitch*:

$f(d,c,b,a)=\sum_4(1,3,4,5,6,7,9,11,12,13,14,15)$

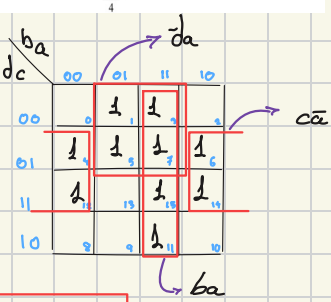


$ii)\ f(d,c,b,a)=\prod_4(0,1,5,7,9,11,12,13)$



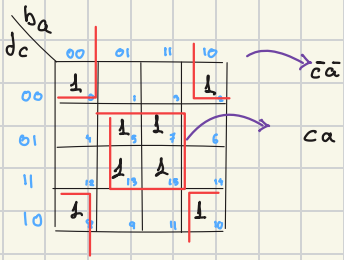
$(\bar{d}+\bar{c}+\bar{b})(\bar{d}+c+a)(d+\bar{c}+a)(d+c+\bar{b})(\bar{b}+a)$

$iii)\ f(d,c,b,a)=\sum_4(1,3,4,5,6,7,11,12,14,15)$



$\bar{d}a+c\bar{a}+ba$

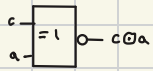
$iv)\ f(d,c,b,a)=\sum_4(0,2,5,7,8,10,13,15)$



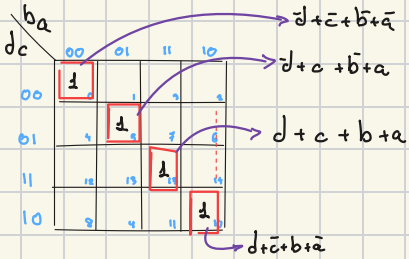
$(ca)+(\bar{c}\bar{a})$

XNOR:

$c\odot a$

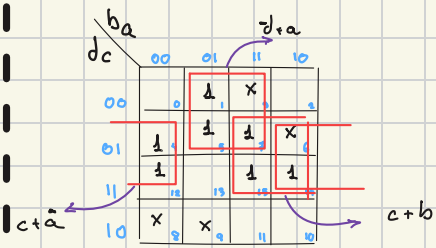


$v)\ f(d,c,b,a)=\prod_4(0,5,10,15)$



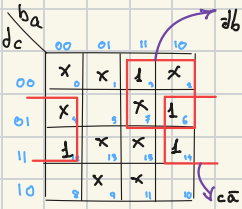
$(d+c+b+a)(d+\bar{c}+b+\bar{a})(d+\bar{c}+\bar{b}+\bar{a})(d+c+\bar{b}+a)$

$vi)\ f(d,c,b,a)=\prod_4(1,4,5,7,12,14,15)\cdot\prod_\phi(3,6,8,9)$

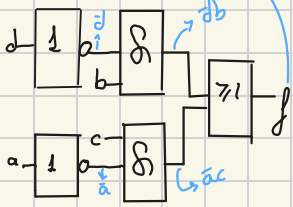


$(\bar{d}+a)(c+\bar{a})(c+b)$

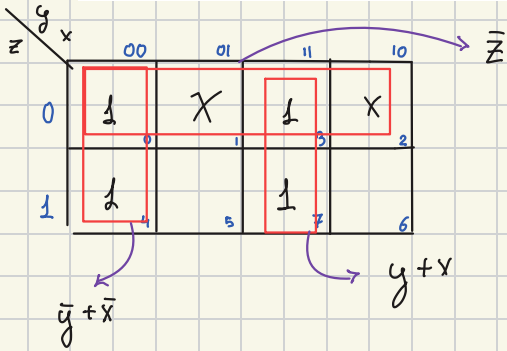
$vii)\ f(d,c,b,a)=\sum_1(3,6,12,14)+\sum_\phi(0,1,2,4,7,9,11,13,15)$



$\bar{d}b+c\bar{a}$

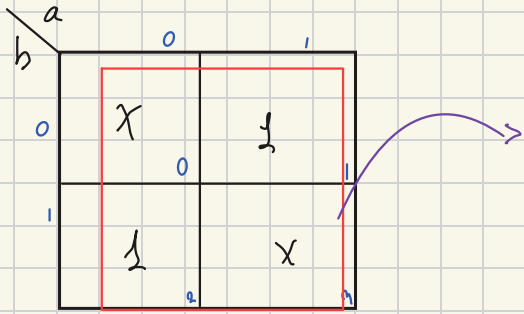


$$viii)\quad f(z,y,x)=\prod_3(0,3,4,7)\cdot\prod_\phi(1,2)$$



$$(y+x)(\bar{y}+\bar{x})\,\bar{z}$$

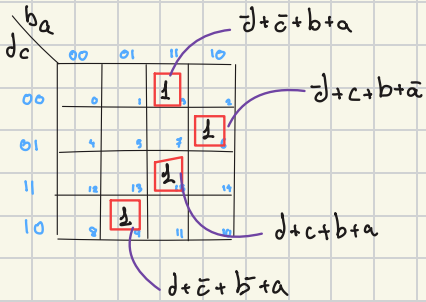
$$ix)\quad f(b,a)=\prod_2(1,2)\cdot\prod_\phi(0,3)$$



Es una cte. siempre vale 1

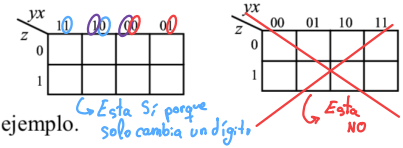
Ejercicio del temario (para comprobar resultados):

$\Pi_4(3,6,9,10,15)$:

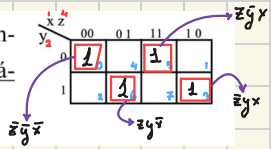


$$\boxed{(\bar{d}+\bar{c}+b+a)(\bar{d}+c+b+\bar{a})(d+c+b+a)(d+\bar{c}+\bar{b}+a)}$$

2) *i)* ¿En tu opinión se puede utilizar alguna de las tablas indicadas en la parte derecha para simplificar funciones lógicas dependientes de 3 variables, con el método de *Karnaugh-Veitch*?. Si se puede utilizar alguna de ellas, indica cual y porqué se puede utilizar. Pon un ejemplo.



ii) Utiliza la tabla de *Karnaugh-Veitch* indicada en la parte derecha para simplificar la función $f(z,y,x)=\sum_3(0,3,5,6)$ e impleméntala utilizando un máximo de 2 puertas lógicas básicas.



$(\bar{z}\bar{y}\bar{x})+(zy\bar{x})+(z\bar{y}x)+(\bar{z}yx)=$

$z\cdot(\underbrace{y\bar{x}+\bar{y}x}_{XOR})+(\bar{z}\bar{y}\bar{x})+(\bar{z}yx)=$

$z\cdot(y\oplus x)+\bar{z}\cdot(\underbrace{\bar{y}\bar{x}+yx}_{XNOR})=$

$z\cdot(y\oplus x)+\bar{z}\cdot(y\odot x)$

xor negado: $(\overline{y\oplus x})$

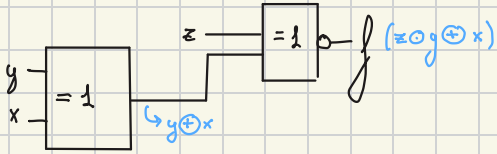
$z\cdot(y\oplus x)+\bar{z}\cdot(\overline{y\oplus x})=$

$\boxed{z\odot(y\oplus x)}$

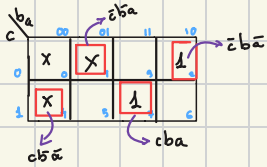
$\underbrace{ab+\bar{a}\bar{b}}_{XNOR}$

z $y\oplus x$

$z\odot y\oplus x$:



4) Hay quien afirma que no eres capaz de implementar la funcion $f(c,b,a)=\sum_3(2,7)+\sum_\phi(0,1,4)$ utilizando únicamente 1 puerta lógica. ¡Demuéstrale que se equivoca!



$c\bar{b}\bar{a}+cba+\bar{c}\bar{b}a+\bar{c}b\bar{a}=$

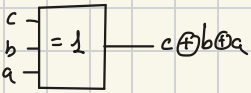
$c\cdot(\bar{b}\bar{a}+ba)+\bar{c}(\bar{b}a+b\bar{a})=$

$c(b\odot a)+\bar{c}(a\oplus b)=$

$c(\overline{b\oplus a})+\bar{c}(b\oplus a)$

$x\bar{y}+\bar{x}y$ (XOR):

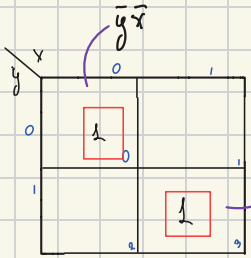
$c\oplus b\oplus a$ → XOR de 3 entradas (no es de las básicas pero es una única puerta)



5) Implementa las siguientes funciones utilizando el menor número posible de puertas logicas básicas.

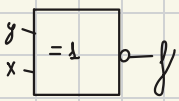
i) $f(y,x)=\sum_2(0,3)$

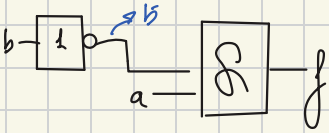
ii) $f(a,b)=\bar{b}+ba$ * $=\bar{b}a$



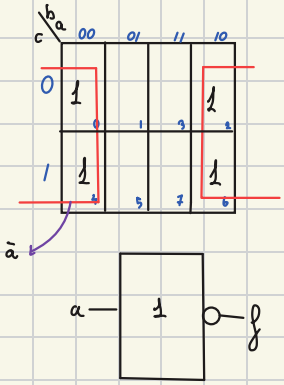
$\underbrace{(yx)+(\bar{y}\bar{x})}_{XNOR}$

$y\odot x$

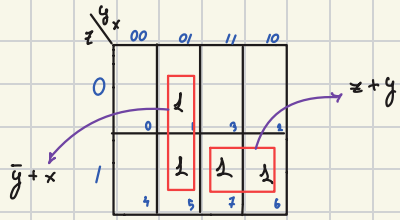




iii) $f(c,b,a)=\prod_3(0,2,4,6)$

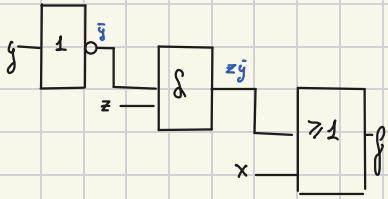


iv) $f(z,y,x)=\prod_3(1,5,6,7)$



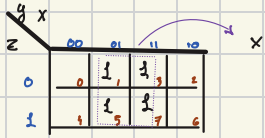
$(z+y)(\bar y+x)$

$z\bar y+\cancel{y\bar y}+x \rightarrow z\bar y+x$



v) $f(z,y,x)=\sum_3(1,3,5,7)$

$f(z,y,x)=\Sigma_3(1,3,5,7)$



"X"

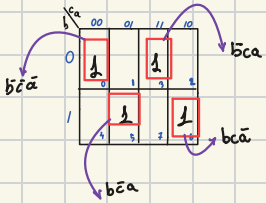
vi) $f(b,c,a)=\prod_3(0,3,5,6)$ *

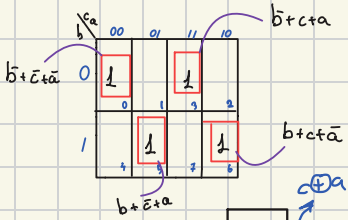
$\bar f(b,c,a)=\prod_3(1,2,4,7)$

$(\bar b+\bar c+\bar a)(\bar b+c+a)(b+\bar c+a)(b+c+\bar a)$

$\overline{\bar f(b,c,a)}=\overline{S_1\cdot S_2\cdot S_4\cdot S_7}=\bar S_1+\bar S_2+\bar S_4+\bar S_7=$

$P_6+P_5+P_3+P_0=\Sigma_3(0,3,5,6)$



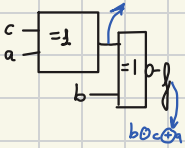


$\bar b\bar c\bar a+\bar b ca+bc\bar a+b\bar c a=$

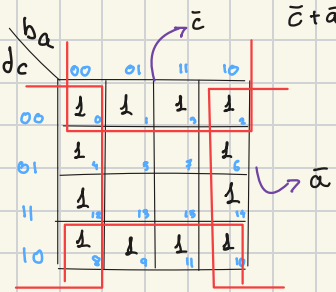
$\bar b(\underbrace{\bar c\bar a+ca}_{\text{XNOR}})+b(c\bar a+\bar c a)$

$\bar b(c\odot a)+b(c\oplus a)$

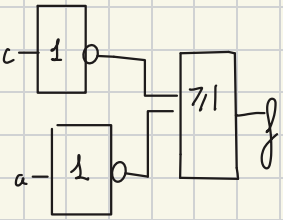
$\bar b(\overline{c\oplus a})+b(c\oplus a)=b\odot c\oplus a$



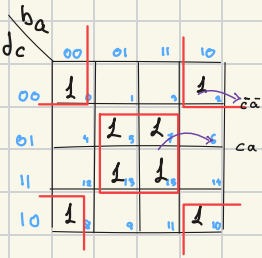
vii) $f(d,c,b,a)=\sum_4(0,1,2,3,4,6,8,9,10,11,12,14)$

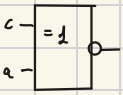


$\bar c+\bar a$



viii) $f(d,c,b,a)=\prod_4(0,2,5,7,8,10,13,15)$



$(ca)+(\bar c\bar a)$

XNOR: $c\odot a$